# Title

Your name here

February 3, 2026

## Exercise 1 [2]

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Tools and resources used: On Exercise 1, I used ChatGPT to help me figure out how to [...]. On Choice Exercise 7, I watched a YouTube video to [...].

Exercise 1 Points attempted: 2. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 2. Questions/Feedback: None.

Exercise 2 Points attempted: 1. Time spent: 2 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.

Exercise 3 Points attempted: 6. Time spent: 40 minutes. Difficulty: 3/10. Self-assessed score: 6. Questions/Feedback: Did I do the problem correctly and as intended?

Exercise 4 Points attempted: 5. Time spent: 35 minutes. Difficulty: 4/10. Self-assessed score: 5. Questions/Feedback: None.

Exercise 5 Points attempted: 1. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.

## Exercise 2 [1]

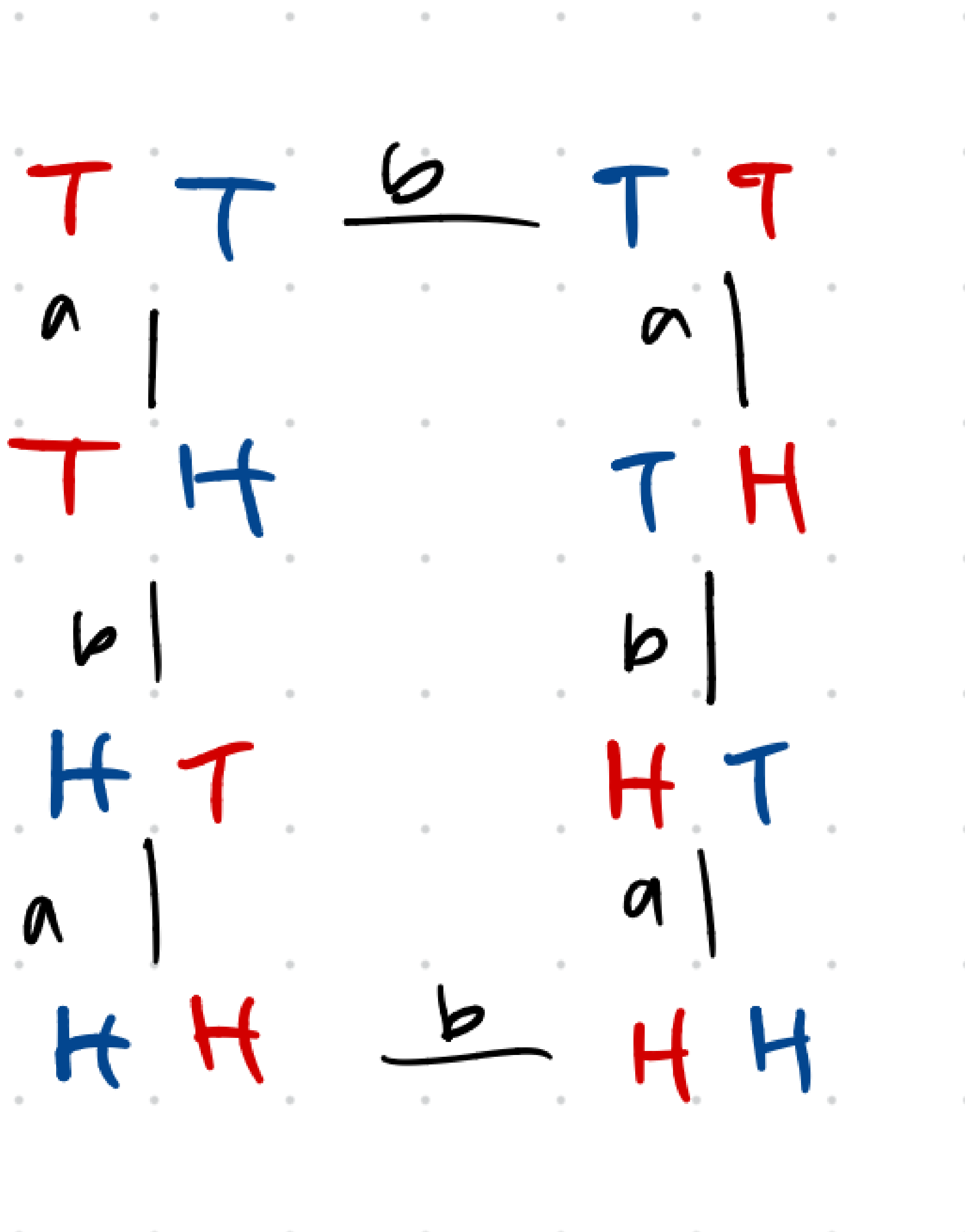
Done! Posted an image of a marker with rotational symmetry.

## Exercise 3 [6]

1. The coins have 8 symmetries. That is because there are $2^2$ configurations without swapping places. When allowing swapping places, we get $2 \times 2^2 = 8$.

2. First, I chose the symmetry of flipping the *left*, denoted as $f_L$, coin over. Applying this symmetry twice gets us back to id. However, when we allow the swapping of places, denoted $s$, we can actually get to all states. We do not actually need an explicit 'flip right' symmetry, since it is actually equal to $f_L \circ s$.

3. In the diagram, let $a = f_L$ and $b = s$.




4. I used 2 generators.

5. This group is not commutative. This is because $f_L \circ s \neq s \circ f_L$. Swapping the coins then flipping the left is not the same as flipping then swapping.

## Exercise 4 [5]

1. First, we write down the numbers in order to see 123456. Now, lets say we want to swap some things around, so we move some things around. Imagine that we move 1 to 2 and 2 to 1. We would get 213456.

If we want to make some more complicated moves, we can actually send any number anywhere! So, $\pi = 152634$ would send 1 to 1 (unchanged), 2 to 5, 3 to 2, 4 to 6, 5 to 3, and 6 to 4.

2. We would get $\pi \circ \pi = 135426$.

3. We would get $(\pi \circ \pi) \circ \pi = 123654$.

4. Since $\pi^3 = 123654$, which is really just swapping 4 and 6, $\pi^3 \circ pi^3 = \pi^6 = 123456$. So, $\pi$ has an order of 6.

5. The right inverse of $\pi$ is $\sigma = 135624$.

6. The left inverse of $\pi$ is $\sigma = 135624$.

7. A word with order 1 is just any of the identity permutations. For example, 1234. For a word with order 2, we have 21. A general formula for constructing $n$ order words is to just have $n$ items, and for the word to be $\pi = 2345...n1$. This will "cycle" the items. So for order 4, we have 2341.

# Exercise 5 [1]

1. The symmetry $r * h$ is a 120 degree rotation followed by a horizontal reflection.

2. The symmetry $r \star h$ is a 120 degree rotation after a horizontal reflection.

3. The function $f \circ g$ behaves like $f$ after $h$.